

Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))

for i in range(len(a)-1):
    m = i
    for j in range(i+1, len(a)):
        if a[j] < a[m]:
            m = j
    a[i], a[m] = a[m], a[i]

print(a)
```

INPUT

```
23 78 45 8 32 56
9 5 1 7
4 2 8 6 1
```

OUTPUT

```
[8, 23, 32, 45, 56, 78]
```

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))

for i in range(1, len(a)):
    key = a[i]
    j = i - 1

    while j ≥ 0 and a[j] > key:
        a[j + 1] = a[j]
        j -= 1

    a[j + 1] = key

print(a)
```

INPUT

```
8 5 7 1 9 3
10 2 8 6
4 1 3 2
```

OUTPUT

```
[1, 3, 5, 7, 8, 9]
```

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))

def merge_sort(a):
    if len(a) ≤ 1:
        return a

    mid = len(a) // 2
    left = merge_sort(a[:mid])
    right = merge_sort(a[mid:])

    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] ≤ right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result += left[i:]
    result += right[j:]
    return result

print(merge_sort(a))
```

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INPUT

38 27 43 3 9 82 10

OUTPUT

[3, 9, 10, 27, 38, 43, 82]

Q4. Diagonal Matrix

Score: 10.00 TC: 3/3 passed

CODE

```
s = input().strip()
s = s[2:-2]

rows = s.split(",")
a = []

for row in rows:
    a.append(list(map(int, row.split(","))))

diagonal = True

for i in range(len(a)):
    for j in range(len(a[i])):
        if i ≠ j and a[i][j] ≠ 0:
            diagonal = False

if diagonal:
    print("Diagonal Matrix")
else:
    print("Not Diagonal Matrix")
```

INPUT

[[1,0,0],[0,5,0],[0,0,8]]

OUTPUT

Diagonal Matrix

Q5. Row and Column Sums

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().replace(" ", "")
s = s.replace("],[", "|")
s = s.replace("[[", "[")
s = s.replace("]]", "]")

rows = s.split("|")
a = []

for x in rows:
    a.append(list(map(int, x.split(","))))

r = [sum(x) for x in a]

c = []
for j in range(len(a[0])):
    total = 0
    for i in range(len(a)):
        total += a[i][j]
    c.append(total)

print("Row=", r)
print("Col=", c)
```

INPUT

[[1,2],[3,4]]

OUTPUTRow= [3, 7]
Col= [4, 6]

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Q6. Maximum Subarray Sum

Score: 6.67 TC: 2/3 passed

CODE

```
a = list(map(int, input().split()))

best = a[0]
current = a[0]

for i in range(1, len(a)):
    current = max(a[i], current + a[i])
    best = max(best, current)

print(best)
```

INPUT

-2 1 -3 4 -1 2 1 -5 4

OUTPUT

6

Q7. Common Elements Among Three Lists

Score: 10.00 TC: 3/3 passed

CODE

```
s = input().strip()
parts = s.split(";")

a = list(map(int, parts[0].split()))
b = list(map(int, parts[1].split()))
c = list(map(int, parts[2].split()))

result = []

for x in a:
    if x in b and x in c and x not in result:
        result.append(x)

print(result)
```

INPUT 1 2 3;2 3 4;2 5

OUTPUT [2]

Q8. Matrix Multiplication

Score: 6.67 TC: 2/3 passed

CODE

```
import json

s = input().strip()
a, b = s.split(" B=")

A = json.loads(a[2:])
B = json.loads(b)

C = []
for i in range(len(A)):
    row = []
    for j in range(len(B[0])):
        total = 0
        for k in range(len(B)):
            total += A[i][k] * B[k][j]
        row.append(total)
    C.append(row)

print("[", end="")
for i in range(len(C)):
    print(C[i], end="")
    if i < len(C) - 1:
        print()
print("]")
```

INPUT A=[[1,2],[3,4]] B=[[5,6],[7,8]]

OUTPUT [[19, 22]
[43, 50]]

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Q9. Sort Rotate Min Max

Score: 0.00 TC: 0/3 passed

CODE

```
s = input().strip()
a, k = s.split(",k=")
a = a.split("=")[1]

lst = list(map(int, a.split()))
k = int(k)

lst.sort()
k = k % len(lst)

rotated = lst[k:] + lst[:k]

print("Sorted= ", lst)
print("Rotated= ", rotated)
print("Min= ", min(lst))
print("Max= ", max(lst))
```

INPUT

List=5 2 8 1 4,k=2

OUTPUT

```
Sorted= [1, 2, 4, 5, 8]
Rotated= [4, 5, 8, 1, 2]
Min= 1
Max= 8
```

Q10. Sort Matrix Search

Score: 0.00 TC: 0/3 passed

CODE

```
s = input().strip()

x = s.split("Matrix=")[1]
mtext = x.split(", Find=")[0]
n = int(x.split(", Find=")[1])

mtext = mtext.replace(" ", "")
mtext = mtext[2:-2]

rows = mtext.split("],[")
m = []

for row in rows:
    m.append(list(map(int, row.split(","))))

r = len(m)
c = len(m[0])

a = []
for row in m:
    for value in row:
        a.append(value)

a.sort()

for i in range(r):
    for j in range(c):
        m[i][j] = a[i * c + j]

out = "Sorted= ["
for i in range(r):
    out += str(m[i])
    if i < r - 1:
        out += ", "
out += "]"

print(out)

found = False
for i in range(r):
    for j in range(c):
        if m[i][j] == n:
            print("Position=", (i, j))
            found = True
            break
    if found:
        break

if not found:
    print("Not Found")
```

INPUT

Matrix=[[9,3],[7,1]], Find=7

OUTPUTSorted= [[1, 3],[7, 9]]
Position= (1, 0)

Q10. Sort Matrix Search

Score: 0.00 TC: 0/3 passed

CODE

```
s = input().strip()

x = s.split("Matrix=")[1]
mtext = x.split(", Find=")[0]
n = int(x.split(", Find=")[1])

mtext = mtext.replace(" ", "")
mtext = mtext[2:-2]

rows = mtext.split("],[")
m = []

for row in rows:
    m.append(list(map(int, row.split(","))))

r = len(m)
c = len(m[0])

a = []
for row in m:
    for value in row:
        a.append(value)

a.sort()

for i in range(r):
    for j in range(c):
        m[i][j] = a[i * c + j]

out = "Sorted= ["
for i in range(r):
    out += str(m[i])
    if i < r - 1:
        out += ", "
out += "]"

print(out)

found = False
for i in range(r):
    for j in range(c):
        if m[i][j] == n:
            print("Position=", (i, j))
            found = True
            break
    if found:
        break

if not found:
    print("Not Found")
```

INPUT

Matrix=[[9,3],[7,1]], Find=7

OUTPUTSorted= [[1, 3],[7, 9]]
Position= (1, 0)